Energy Wave

CONFIDENTIAL — DRAFT — FOR DISCUSSION · NO MEASURED ENERGYWAVE PERFORMANCE EXISTS · LEAD THREAD OPEN (D2)

What a technical reviewer receives with the seed round. Every document, its status, and what it does and does not establish.

The technical hub for the seed round and the companion to the Seed Round Structure page, the Seed Deck and the Seed One-Pager: the technical material a seed investor's reviewer is given, what each item establishes, and the standard every later result is held to. It states no measured EnergyWave performance, because none exists; it says what would have to be measured, and replicated, for Tranche B to be released.

SCOPE STATEMENT — READ BEFORE ANY DOCUMENT IN THIS FOLDER

> No measured EnergyWave hardware performance evidence has been reviewed for this package. Every number in an EnergyWave-authored document is labelled by origin — *assumption · calculation · proposed target · company measurement · attributed published result* — and no EnergyWave document carries a company measurement yet. Figures cited from published research are attributed to their authors and are not EnergyWave results. Every claim the company examines must answer two questions: **what is the net energy source, and where is the colder sink?** The reviewers' refinement is adopted here: a prepared, stored or driven state is itself an answer to the first question and is counted as an input; a colder sink is the test for a heat-engine claim, and the release of a stored potential is accounted as storage, never as generation.

WHAT THE SEED FUNDS, TECHNICALLY

Tranche	Technical deliverable	Evidence that releases the next tranche
A released at close	Principal investigator hired (a condition of the tranche). Thread C: the artifact-exclusion bench commissioned on known-null samples; its error budget published before any result is claimed (Error Budget page). Thread B: the pre-registered replication of Moddel et al. 2021 — cavity / no-cavity twins from the same wafer, blinded measurement, sign and magnitude against cavity thickness. Thread A: the founder's solid-state module characterized under exactly the same protocol, once the inventor-defined inputs are on record (Thread A plan; register B-4; D2).	Gate-1 report. Pass: repeatable output above the fully characterized noise and artifact floor at ≥5σ in blinded runs, surviving every control. Kill: nothing above the floor, or output explained by a known source — published with data. Either outcome is a result; only the first releases Tranche B.
B escrowed	Gate 2 — closed-system calorimetry; mechanism discrimination against known-source models (finite stored-energy depletion, open-system environmental transduction, metrology phase skew); independent replication at a second laboratory contracted before Gate 2 begins.	Gate-2 report: net output whose closed-system energy balance cannot be closed by known sources, replicated by the second laboratory. Otherwise the effect is retired publicly.

Tranche	Technical deliverable	Evidence that releases the next tranche
C escrowed	Gate 3 — engineering characterization, lifetime and stability, third-party validation under observation; the anti-tamper “paperweight test”.	The first point at which EnergyWave would ever state a performance number. Scaling, lifetime or stability failure ends the program having answered the question.

Tranche amounts, the seed amount and the instrument are the founder’s decisions with counsel and are tagged in the Seed Deck and One-Pager (register B-2); the raise model’s stored figures are placeholder outputs, never approved targets.

ENERGY ACCOUNTING — THE STANDARD EVERY THREAD IS HELD TO

> **$E_{\text{all inputs}} = E_{\text{useful load}} + E_{\text{other leaving}} + \Delta U_{\text{device}}$**, where U includes everything stored in capacitors, batteries, magnetic and strain states and internal thermal reservoirs. Preparation, drive, control, cooling and reset energy are counted once each; a reset already inside a full-cycle measurement is not counted twice. DC input is ∫v(t)i(t)dt, never average V × average I; probe deskew in a high-Q circuit can turn a true +50 mW dissipation into a false −75 mW “generation” reading at 1 MHz. A null result bounds the tested configuration at its demonstrated sensitivity — it does not identify an origin. A voltage recovery, a resonant amplification ratio, a short-circuit current trace or a single discharge is an observation, not a useful-load energy measurement.

THE TECHNICAL DOCUMENT SET

Document	Status	What it establishes — and does not
Research Program and Gates (2026-10-15) 03_Technical_Documents	DRAFT	The research question, the three-gate protocol with pass / kill criteria, the energy-accounting standard, threads A / B / C under one protocol with no dependency asserted, the eight commitments. The lead thread is the founder’s decision (D2).
Thread A Characterization Plan (2026-10-15) 03_Technical_Documents	DRAFT	The first testable question for the founder’s module; hypotheses H-A1 stored-energy depletion, H-A2 environmental transduction, H-A3 metrology phase skew; off-the-shelf baseline proposals; the four-step unconditioned characterization; pre-registered pass / kill / ambiguous rule. The specimen, magnetization, winding axis, conditioning procedure and the founder’s notes are inventor-defined and tagged (register B-4).
Measurement Platform Error Budget (2026-10-15) 03_Technical_Documents	DRAFT	Thread C for external review: the measurand (output power into a defined load), the error-budget table with every term “to be measured” until commissioning, the commissioning sequence on known-null samples, calorimetry, sign-off. The ≤ 0.14 pW floor is a *proposed target* inherited from the prospectus, ratified at commissioning; effect size, resolution and leads tagged (register B-6).
Humanoid Power-Budget Specification (2026-10-15) 03_Technical_Documents	DRAFT	Capital-readiness objective #2: method, attributed platform benchmarks, the six-row subsystem worksheet and the specification outputs ($\bar{P}$, P_{peak}, ramp, mass, volume, heat rejection). Platform inputs tagged (register B-5). It defines the question a power source would have to answer; it asserts nothing about any EnergyWave device.

Document	Status	What it establishes — and does not
Physical Architecture & Baseline Model v1.3 (2026-10-05) 03_Technical_Documents · 6 pp	ORIGINAL	AI-reviewed working briefing: reservoir inventory, energy accounting, metrology, the three conditioning hypotheses, an unconditioned baseline characterization before any conditioning experiment. §6 is a historical legal table superseded by 04_Legal_and_IP; its “63/763,986” reference is unsupported. Canonical build to be confirmed by Ian (D8). Reference, never a submission document.
Technical Prospectus v1 (2026-09-07) 03_Technical_Documents · DOCX	HISTORICAL	The September program in full: §4 physics with labels; §6 Gate-1 design and artifact catalogue; §6.5 workstream 1C — the founder’s own structures under the same protocol, with the conditional note that Forward’s charged foliated conductor is a one-shot store; §6.6 decision rule. Carries bracketed placeholders; reference only.
Five technical notes of 2026-10-05 03_Technical_Documents · Markdown	ORIGINAL	Claude’s engineering review v2 of the briefing; Puthoff & Little assessed (best-case single-use Casimir stack ≈ 2 J/kg against ~9×10^5 J/kg for lithium-ion — *calculation*); regauging essay, Pais patents and Exodus Propulsion checked; Antigravity’s GR-coupling evaluation and its review (the arithmetic holds; no measurable metric effect at a 1 µm gap). AI-authored; labelled as such.
Key Research Facts (2026-09-07)	HISTORICAL	One-page corrections to common belief; the energy figures used in the narrative.
Claims Discipline (2026-10-15) package root	DRAFT	The rules every document obeys — two-question filter, evidence labels, no device figures, publish both ways — and the recorded difference between the September charter and the Oct-13 deck (D10).
xTech Search 10 white paper — as submitted (2026-10-09) and the approved replacement; DoW White Paper v2.1 (draft) 02_DoW	ORIGINAL DRAFT	Thread C as proposed to the Army: MIM-diode / Casimir-cavity metrology as an independent evaluation platform; the submitted cover states TRL 2. A submission is not an award and not a validation. The v2.1 pass limits “unbuilt theoretical estimates” to kilowatt-scale extraction figures and states Moddel’s picowatt-scale result as disputed and not independently replicated.
Compartmentalization & Anti-Tamper Posture v0.1 (2026-09-12) 02_DoW	ORIGINAL	How a positive result would be protected (CPI / AT). A v0.1 working memo whose §8 counsel questions are open; the working canonical among three variants is to be confirmed (D8).
Literature cross-check of the eight archive PDFs (2026-10-14) not placed — EnergyWave Final / Literature / Review; ledger S-37, S-42	REPORTED	GPT/Lyra’s reading map and three detailed reports (all 28 Puthoff-group pages and Forward’s 3 pages read in full; Cook, Utkin, Nair and the AI transcript text-screened with recorded samples), with Antigravity’s and Gemini/Argon’s independent responses and Lyra’s correction notes on each. All three agree: legitimate physics and useful history; no repeatable vacuum-powered engine, no validated ferrite mechanism, no measured EnergyWave performance. Secondary AI evidence, not replication; the findings are carried into the table below.

WHAT THE EVIDENCE ESTABLISHES TODAY

Claim or source	Status	Basis, as the sources state it
A repeatable EnergyWave output of any kind	OPEN	None measured. No specimen selected, no hardware ordered, no result. Thread A's specimen, magnetization, winding axis and conditioning procedure are inventor-defined and not yet on record (register B-4).
Moddel et al. 2021, optical-cavity-induced current (MIM diode under a Casimir cavity)	CLAIMED, NOT REPLICATED	*Attributed published result.* A maximum of 1.4 pW inferred from the I–V curve for one device geometry; repeatability across more than 1,000 devices in 21 batches in the same laboratory. The reported four-hour trace is short-circuit current, not metered useful-load energy. Vacuum excitation is the authors' proposed interpretation. Maclay 2024 argues the cavity model predicts the wrong current direction and the wrong cavity-size behaviour — a critique of mechanism, not a replication. No independent replication establishing vacuum origin and a complete sustained net-output system was located in the reviewers' bounded search.
Forward 1984, charged foliated conductors	THEORY	A proposed rechargeable Casimir battery. Re-expansion of the plates explicitly costs energy from the bidirectional supply (p. 3). A finite discharge with a paid reset: the cycle is non-positive, not a repeating net-power engine. Carried as a one-shot store in Prospectus §6.5.
Puthoff & Little (JBIS 55, 2002; arXiv 2010); Puthoff, Phys. Rev. D 35 (1987); Davis et al., STAIF 2006	THEORY / PROPOSALS	Exploratory theory and a proposed experimental program. The 1987 result is an equilibrium balance for a restricted circular orbit (footnote 17 limits it), not surplus power for a load. The 2006 review reports an inconclusive cavity / synchrotron experiment and a switchable-mirror estimate that assumes negligible switching energy; it reports no practicable demonstration as of 2006, which is not a measured universal zero. The local two-author Puthoff & Little record is distinct from the withdrawn three-author record.
The Casimir force	MEASURED	Adjacent science (Lamoreaux 1997, torsion-pendulum force measurement). The force is conservative: whatever work it does closing the plates must be paid back to open them. A measured force is not a generator.
Best-case single-use Casimir stack vs. lithium-ion	CALCULATION	≈ 2 J/kg against ~9×10^5 J/kg for the lithium-ion cell a humanoid already carries (Claude source note, 5 Oct). Said here before an investor's physicist friend says it.

Claim or source	Status	Basis, as the sources state it
Cook 2002; Utkin 2012; Nair 2021; a ChatGPT conversation export of 2024	BACKGROUND	Not for any claim. Cook p. 122 reports no demonstrated excess output among ~30 devices; p. 298 orders giga and tera wrongly and compares power to energy. Utkin pp. 54–61 contain preliminary bench observations without a closed energy inventory; a control without the shunt diodes gives a similar trace. Nair Eq. 17 takes the square root of an energy density and keeps energy-density units. The transcript's p. 20 formula adds terms of incompatible dimensions. Preserved as history; cited for nothing.

DRAFTED AND WAITING ON INPUTS; STILL TO WRITE

Item	Owner	Gate
Thread A characterization plan — drafted; specimen, magnetization, winding axis, conditioning procedure, the founder's notes and prior measurements	Claude with Chip Story	Register B-4; D2 OPEN
Humanoid power-budget specification — drafted; platform, mass and bay volume, duty profile, heat rejection, the six subsystem rows	Claude	Register B-5 OPEN
Measurement-platform error budget — drafted; the thread A effect size to resolve, the calorimetric resolution to fund, the thread C and measurement leads	Thread C lead / PI	Register B-6 OPEN
Pre-registration template, blinding procedure and open-data pipeline specification — to write	PI	Before the first Gate-1 run OPEN
Second-laboratory replication terms, contracted before Gate 2 begins — to write	PI with counsel	Gate 2 OPEN
Scientific advisory board charter with veto over any performance statement; at least one declared critic — to write	Ian	D4 — consents on file OPEN

OPERATING REQUIREMENTS — FACILITIES, BENCH, STAFFING (REGISTER PENDING)

Ian has directed that this folder also serve as the technical seed hub — laboratory and workbench, an office near UC Irvine, staged staffing and every other seed requirement, with evidence (relayed by GPT/Codex, 15 Oct — S-43). GPT/Codex is preparing a source-backed *operating-requirements and quote register*; when received it is placed as its own dated file beside this page and this table is filled from it. Until then no facility, bench, headcount or cost figure is stated. **[Needs Input from Ian or Chip: confirm the target site near UC Irvine, the laboratory footprint and the first two hires once the operating-requirements register is received]**

Register section	What it must carry	Evidence rule
Facilities	Office and laboratory near UC Irvine: candidate spaces, size, services (power, HVAC, vibration, shielding), lease terms, dates.	Dated quote or listing, vendor named OPEN
Minimum bench (thread C)	The artifact-exclusion bench as a parts list: shielded isothermal enclosure, calorimetry, lock-in electrical measurement to the picowatt level, probes and deskew calibration, blinded sample handling, data pipeline — the Error Budget page's terms, costed.	Dated vendor quote per line; origin label *proposed target* until purchased OPEN

Register section	What it must carry	Evidence rule
Staged staffing	Principal investigator at Tranche A; technician; second-laboratory contract at Gate 2; advisory board. Roles, start month by tranche, consents.	Role described; no named person without written consent (D4) OPEN
Budget evidence	Each line of the raise model's UseOfFunds that the register validates or replaces, with the quote behind it.	v1 workbook figures stay historical placeholders; no raise amount is derived here (D1; register B-2) OPEN
Milestones	Facility ready, bench commissioned on known-null samples, error budget published, first pre-registered run — each tied to a gate month.	Dates are *proposed targets* until the PI signs them OPEN

CROSS-CHECK STATUS FOR THIS PAGE

> This page is a draft until GPT/Codex and Antigravity have returned findings on it in the one-line format of REVIEW_PROTOCOL.md (register R-1) and Ian has answered D2. A finding of *blocker* or *major* re-renders the page; the PDF is never edited directly. Grok reviews any outbound draft for compliance (none sent). Gemini/Argon's completed literature response carries Lyra's seven qualifications (S-37) and is not a package review.

Reviewer	Reviewed	Pending
GPT / Codex (Lyra / Aura)	r3 audit (S-31); r5 submission-readiness audit, r5.2 delta and r5.3 closure (S-39–S-41). Literature reading map and three detailed reports, 14 Oct (S-37; read in full for this page, S-42).	Delta check of this release, including this page. The operating-requirements register (S-43).
Antigravity	Independent r3 review (S-32); GR-coupling evaluation of 5 Oct (S-17); independent literature review of 14 Oct, recorded with Lyra's ten corrections (S-37).	Independent review of this page after GPT/Codex, as Ian requested.

Sources: Seed Round Structure, Research Program, Thread A Plan, Error Budget and Power-Budget pages (this package); deck slides 4–7, 9 (S-01); Technical Prospectus v1 §§4, 6.5–6.6 (M-05); Physical Architecture & Baseline Model v1.3 §§1–3 (M-14 / S-11); Claude review v2 §3 and source notes (S-13, S-15–S-18); xTech papers (M-10, M-06); AT posture v0.1 (M-13); literature cross-check (S-37) and the GPT/Codex detailed reports (S-42); Lyra coordination note, 15 Oct (S-43); GPT/Lyra r5 audits (S-39–S-41); Antigravity r3 review (S-32); DECISIONS.md D2, D4, D8; register B-2, B-4–B-6, B-12.